TOPOLOGY / UNKNOT RECOGNITION

Single-Exponential Activation Kernels

for Dual-Forest Selection

Generic mandatory contraction, representative families,
and deterministically checkable low-rank obstructions

Research contribution prepared with ChatGPT
for Vladimir Reshetnikov's *ProveIt* project
12 October 2026

Abstract

The audited incoming ProveIt dual-forest selector has a deterministic $2^{O(b\log(b+1))}\,\mathrm{poly}(\Lambda)$ bound, where b counts rows incident to usable higher-support columns and Λ is the encoded local input size. We give a one-sided randomized $2^{O(b)}\,\mathrm{poly}(\Lambda,\lambda)$ search with miss probability at most $2^{-\lambda}$ on positive inputs. Every reported success is an independently replayable, entirely discrete certificate. The proof replaces exceptional-row permutations and shared-witness partitions by a closed activation state, at most b useful additional seeds, generic contraction of mandatory color representatives, and exterior-algebra representative families. Only the dual graph is represented over a finite field; geometric intersection counts are never reduced modulo that field.

When there are no mandatory low-column source components and the dual graphic rank is at most b, the same method is deterministic and exact. A negative answer has a frontier-span certificate checked from source graph minors and literal peeling, without trusting the optimizer. We also give a necessary source-chain parity condition and a polynomial criterion for parity-compatible lifts of saturated intersection data.

The package contains 71 passing unit tests, 8,061 exhaustive/random matrix–graph comparisons, 300 planted positive trials, independent finite-field audits, comparative timings, and an opt-in record-level native adapter. All 3,417 positive trials were certified; 3,391 negative inputs had exact local certificates and 1,553 remained explicitly inconclusive. These results improve a specified local selection stage. They do not establish a general quasi-polynomial unknot recognizer or a geometric realization theorem for the abstract benchmark families.

Status. Conventional proofs and executed research code; not externally refereed or proof-assistant formalized. The contribution is measured against the audited predecessor, not claimed as a historically first use of representative families. Two other recently incoming archives were identified but were not content-audited.

Contents

1 Research position and the advance **1**
1.1 The checkpoint being continued . . . 1
1.2 What changes, and what does not . . . 1

2 The fixed cancellation problem **2**
2.1 Unsigned counts and dual edges . . . 2
2.2 Why static peeling is the correct local target . . . 2
2.3 The geometric interface is an additional hypothesis . . . 3

3 Activation closure and the small seed lemma **3**
3.1 Low-column implications . . . 3
3.2 A closure operator only on the exceptional boundary . . . 4

4 Mandatory choices as a generic contraction **5**
4.1 The graph representation . . . 5
4.2 Specialize, quotient, and reduce the rank . . . 6
4.3 A witness can be unmixed deterministically . . . 7

5 Exterior representative families over closed states **7**
5.1 The representative property . . . 7
5.2 The dynamic program . . . 8
5.3 A representation invariant that matters in code . . . 8

6 The single-exponential search theorem **8**
6.1 Why only one witness has to survive randomness . . . 8

7 Deterministic low-rank decisions and frontier certificates **10**
7.1 A class with no randomness at all . . . 10
7.2 What the negative packet contains . . . 10

8 Necessary parity checks for geometric source data **11**

9 The maintained loop example and the integration seam **12**

10 Implementation, resource semantics, and certificates **13**
10.1 Modules and independent paths . . . 13
10.2 The three statuses . . . 14
10.3 Reproduction commands . . . 14

11 Executed validation and measured tradeoffs **14**

11.1 A ledger that distinguishes unknowns from negatives 14
11.2 An exact finite-field probability check 15
11.3 A nonlinear family with exact witness-count separation 15
11.4 Paired local-query timings 16

12 What this contributes to a quasi-polynomial recognizer **17**

13 Further research questions and concrete next targets **18**
13.1 Deterministic single-exponential mandatory contraction 18
13.2 Source restrictions stronger than chain parity 18
13.3 A uniform geometric bound on the activation boundary 19
13.4 Joint optimization of primal and dual forests 19
13.5 More exact negative routes without full symbolic expansion 19
13.6 Fewer exceptional bits and stronger safe preprocessing 19
13.7 Weighted objectives and faster exterior computation 19
13.8 Native replay, proof formalization, and adaptive sampling 19

14 Conclusion **20**

A The classical mandatory-representative oracle **20**

B Source audit and contribution ledger **21**

1 Research position and the advance

1.1 The checkpoint being continued

This study fixes the maintained repository at commit

`e0b2f802f336fcb51e9ec5f29174cfe28dfb9d39`.

The source handles, dual endpoints, preservation of bundle ports, and the five-row loop experiment are taken from the inspected maintained materials [1, 3, 4, 5]. The immediate algorithmic predecessor is *Matroidal Dual-Forest Selection* [6], supplied in the incoming archive of 12 October 2026. Its higher-support algorithm considers exceptional row orders and partitions of the demands that may share a seed. The bound is $b!\,\mathrm{Bell}(b)\,\mathrm{poly}(\Lambda)$, hence $2^{O(b\log(b+1))}\,\mathrm{poly}(\Lambda)$. Its question on replacing this dependence by $2^{O(b)}$ is the concrete target addressed here.

The distinction between choosing *some* boundary-rooted dual forest and choosing a *useful* one is genuine. In the maintained loop example, the default tree deletes a row that leaves a stopping set, while three alternative choices allow completion [4]. The prior selector already resolves that coverage issue. Our reconstructed loop experiment is consequently a performance and certificate-replay comparison, not a claim to have newly covered that source.

The ingredients from linear matroid representative-family theory are classical; useful algorithmic accounts include [9, 10]. Our contribution is the precise reduction of this activation problem to a small representative-state search, including how mandatory source components are handled without retaining all their choices. Generic linear combinations are used only as a search device. The final tree and peeling schedule do not contain field elements or symbolic variables.

1.2 What changes, and what does not

Write $r = |V(D)| - 1$ for the rank of the connected dual graph, and let q be the number of mandatory components defined in Section 3. After an optional exact mandatory-obstruction prepass, put

$$d = \min\{b, r - q\}.$$

One random round has the conservative field-arithmetic bound

$$\mathrm{poly}(\Lambda) + 2^b 8^d\,\mathrm{poly}(\Lambda) \;\le\; 16^b\,\mathrm{poly}(\Lambda), \tag{1}$$

with single-exponential bit complexity as well. Repeating λ times makes the miss probability at most $2^{-\lambda}$ on any input that has a successful tree. The dependence on the number q of mandatory components is polynomial, not exponential. When $q = 0$ and $r \le b$, the original binary incidence representation is already small enough: the search is deterministic and can certify either outcome.

The input remains a fixed unsigned core matrix and a fixed dual graph, with the original zero/one-handle forest frozen. The method does not optimize that primal forest, discover a normal surface, establish admissibility, or maintain all boundary-pattern data through an entire hierarchy. A negative local certificate says that *this particular tree-plus-positive-cancellation strategy* cannot succeed. It says nothing by itself about whether the source is a handlebody or whether a knot is trivial.

Lackenby's announced target is $n^{O(\log n)}$ in the crossing number [7]. The inspected 2026 hierarchy paper distinguishes its iteration accounting from the still unspecified costs of some operations in Section 9 [8]. The local quasi-polynomial regime proved below is therefore stated with explicit source-size, call-count, and feasibility hypotheses, rather than presented as a complete recognition bound.

2 The fixed cancellation problem

2.1 Unsigned counts and dual edges

Let $E = \{0, \dots, m-1\}$ be a row set, let C be a column set, and let

$$N = (n_{eh})_{e \in E, h \in C} \in \mathbb{Z}_{\geq 0}^{E \times C}.$$

Rows model source two-handles, and columns model the one-handles remaining after the chosen zero/one forest contraction. Counts are unsigned geometric intersections when a geometric source is supplied. For the abstract selection problem only the categories 0, 1, and at least 2 matter.

Let $D = (V, E)$ be a connected multigraph with a distinguished boundary root 0. The same label e denotes a matrix row and a dual edge. Parallel edges and loops are retained; a loop is not a forest edge. Let $r = |V| - 1$. A spanning tree T first deletes its row set from N. Thereafter a row e can be peeled using column h exactly when

$$n_{eh} = 1, \qquad n_{fh} = 0 \quad \text{for every other live row } f. \tag{2}$$

Both the row and that pivot column are then removed. A *successful tree* is a spanning tree after whose deletion all remaining rows can be peeled.

The encoded input size Λ includes the graph, column information, matrix entries, and the bit lengths of the supplied counts. All graph and matrix objects used here are explicit. The implementation retains original counts for input binding, while tests of $n = 0$ and $n = 1$ implicitly saturate larger counts. Nothing is proportional to the numerical value of a large intersection count.

2.2 Why static peeling is the correct local target

The maintained cancellation theorem and the audited predecessor show that private-column peeling is complete for the positive unsigned update model [2, 6]. We recall the stopping-set argument so that the local criterion is unambiguous.

Definition 2.1. A nonempty set of rows R is a *stopping set* if, for every column h,

$$S_h(R) := \sum_{e \in R} n_{eh} \quad \text{is either 0 or at least 2.}$$

A fixed-row deletion set succeeds under peeling if and only if it meets every stopping set of the original matrix. Indeed, a peeling process cannot remove the first row of a surviving stopping set. Conversely, the nonempty terminal set of any stalled peeling process is a stopping set. Thus the terminal set is independent of the legal peeling order: it is the union of all stopping sets avoiding the initial deletions.

For completeness with respect to the positive-fill model, an unsaturated unit pivot at (e, g) performs

$$n'_{fh} = n_{fh} + n_{fg} n_{eh} \qquad (f \neq e,\ h \neq g).$$

If $e \in R$ and $n_{eg} = 1$, then

$$S'_h(R \setminus \{e\}) = S_h(R) + (S_g(R) - 2) n_{eh}. \tag{3}$$

Since $S_g(R) \geq 2$, no surviving stopping-set column acquires total one. The remaining row set is nonempty because e contributes only one to column g. A pivot outside R cannot use a column of total zero on R to create an entry there, and otherwise only increases its entries. Stopping sets therefore

persist through arbitrary positive pivots. Saturating each entry at two preserves precisely the distinction between a column total of zero, one, and at least two. This proves that a stalled private-column process cannot be repaired merely by choosing other positive-fill pivots.

This argument does *not* apply to arbitrary signed changes of attaching words, geometric slides outside the maintained positive model, or changes of the primal forest. Such moves may be useful, but are different operations.

Lemma 2.2 (Monotonicity and tree extension)**.** *If row deletion set F permits complete peeling, so does every superset of F. Consequently, a successful graphic-independent deletion set can be extended to a successful spanning tree of D.*

Proof. Follow any successful peeling order, omitting rows already deleted by the larger set. A formerly private unit column remains private whenever its pivot row is still present. A column whose pivot row has already disappeared can simply be omitted. Equivalently, deleting more rows cannot leave an original stopping set intact. Finally every forest in a connected graph extends to a spanning tree, including in a multigraph with loops excluded from the forest. □

2.3 The geometric interface is an additional hypothesis

For the inspected source charts, an extreme edge-gap two-handle has two authenticated dual endpoint occurrences. A source interior vertex contributes a three-handle; original-boundary and cut-boundary endpoints are represented by the root. Contracting a rooted spanning tree gives legal two/three cancellations and preserves the lower zero/one body and bundle port charts [3]. A repeated interior endpoint is two intersections, not one.

Our abstract selector only proposes which such rows to delete. Its graph input does not prove that these endpoints came from an actual source. The maintained checker reconstructs endpoints independently from vertex-link triangles and normal-coordinate edge weights. A proposed native record must pass that checker, followed by the ordinary core, port, boundary-pattern, and whole-cut checks. The article's main theorem is complete for the stated fixed matrix/graph problem, not for all possible downstream geometric objectives.

3 Activation closure and the small seed lemma

3.1 Low-column implications

Call a column *usable* if it has at least one entry equal to one. Columns without a unit entry can never be peeling pivots. Build a directed graph G on the row set E using usable columns of support at most two. A unit singleton is an *anchor*; let A be the set of anchors. For a column with support $\{u, v\}$, include the arc $v \to u$ when $n_{uh} = 1$, and include $u \to v$ when $n_{vh} = 1$. In particular $(1, 2)$ gives a directed implication from the heavy row to the unit row, not an undirected edge.

For $S \subseteq E$, let $\mathrm{Reach}(S)$ include S and all vertices reachable from it in G. Low-column peeling from seed set F deletes exactly $\mathrm{Reach}(A \cup F)$. The forward implication follows by reading a peeling step as an arc; the reverse follows by following directed paths. Once a column is used, all rows in its support are already deleted, so retaining its implication for closure purposes introduces no spurious later move.

Let $\mathcal{H}$ be the usable columns with support at least three, and define

$$B = \bigcup_{h \in \mathcal{H}} \mathrm{supp}(h), \qquad b = |B|. \tag{4}$$

All support rows, including entries at least two, belong to B. Omitting heavy prerequisites is not a valid reduction. By contrast, a high-support column with no unit entry contributes nothing to B.

A source strongly connected component of G is *mandatory* if it contains neither an anchor nor a row of B. Write these components as $U_1, \dots, U_q$. They are pairwise disjoint. Define

$$R_0 = \mathrm{Reach}\left(A \cup \bigcup_{i=1}^{q} U_i\right). \tag{5}$$

The use of entire U_i in this symbolic baseline does not grant their rows free deletions. It is justified only after a genuine seed is chosen in each U_i. Any one row of a strongly connected component reaches all of it and has the same low-column descendants as every other row in the component.

Lemma 3.1 (Mandatory representatives)**.** *Every successful initial deletion set meets each U_i. If it contains one row of every U_i, then all of R_0 can be deleted using anchors and low-column implications.*

Proof. Consider the first row of an unseeded U_i that would be peeled. It cannot be an anchor. A low-column predecessor outside U_i would be an incoming arc to a source component, and a high-column pivot would put its row in B. Both are impossible. Conversely, one seed propagates through its strongly connected component and all its descendants. □

3.2 A closure operator only on the exceptional boundary

For each row e define its influence mask

$$H(e) = B \cap \mathrm{Reach}(\{e\}).$$

For $S \subseteq B$, define $\mathrm{cl}(S)$ as the smallest subset of B containing $S \cup (B \cap R_0)$ and closed under the following two sorts of implications:

$$x \in \mathrm{cl}(S) \implies H(x) \subseteq \mathrm{cl}(S) \qquad (x \in B), \tag{6}$$

$$\mathrm{supp}(h) \setminus \{x\} \subseteq \mathrm{cl}(S) \implies x \in \mathrm{cl}(S) \qquad (h \in \mathcal{H},\ n_{xh} = 1). \tag{7}$$

Every prerequisite of (7) is in B, so this is a finite closure system on b bits. It is extensive, monotone, and idempotent. The baseline $\mathrm{cl}(\varnothing)$ may contain high-column consequences as well as $B \cap R_0$. A direct saturation takes polynomial time per mask, and at most 2^b masks ever need memoization.

Theorem 3.2 (Exact activation reduction)**.** *Let I consist of one representative from each mandatory component, and let $X \subseteq E$ be any additional seed set. The exceptional rows deleted by full peeling from $I \cup X$ are exactly*

$$\mathrm{cl}\left(\bigcup_{e \in X} H(e)\right). \tag{8}$$

In particular full peeling succeeds if and only if this set is B.

Proof. Each generating implication is a valid low-column path or a valid private high-column pivot. Lemma 3.1 supplies the baseline. Thus the right-hand side of (8) is attained by genuine peeling.

For the converse, inspect a full peeling sequence. A seed in X supplies $H(e)$, and the mandatory seeds and anchors supply R_0. Until a high pivot occurs, all newly deleted rows lie in the corresponding low reachability closure. The target and every prerequisite of a high pivot are in B; therefore (7) already accounts for it. Low propagation from that target is accounted for by (6). Induction along the sequence proves equality of the exceptional projections.

If all rows are deleted then certainly all of B is deleted. Conversely, every source component of G either has an anchor, is mandatory, or meets B. Once B is deleted, each source component has a deleted vertex. Every component of the finite condensation graph lies downstream of a source. Low-column propagation therefore deletes all rows, including rows outside B. □

Theorem 3.3 (At most b useful additional seeds)**.** *A successful spanning tree exists if and only if there are a set X with $|X| \le b$ and representatives $u_i \in U_i$ such that*

$$X \cup \{u_1, \dots, u_q\} \text{ is a forest}, \qquad \mathrm{cl}\left(\bigcup_{e \in X} H(e)\right) = B. \tag{9}$$

The representatives and X can be chosen disjoint. Furthermore X has an ordering in which every added seed strictly increases the closed state.

Proof. Take a successful tree T. Choose one representative $u_i \in T \cap U_i$ for every mandatory component, which is possible by Lemma 3.1. Start at $S = \mathrm{cl}(\varnothing)$ and repeatedly choose an edge e of T for which $\mathrm{cl}(S \cup H(e)) \neq S$.

If this process stopped at a proper subset of B, every $H(e)$ for $e \in T$ would already be contained in S. Idempotence would then give $\mathrm{cl}(\bigcup_{e \in T} H(e)) = S$, contradicting successful peeling and Theorem 3.2. Every chosen edge adds at least one previously absent bit, so at most b edges are chosen. No row in a mandatory component can be chosen: its influence is already contained in $B \cap R_0$. Hence X and the representatives are disjoint, and their union is a subset of T, thus a forest.

Conversely, (9) and Theorem 3.2 give a successful forest deletion set. Lemma 2.2 extends it to a successful tree. The greedy construction gives the claimed strict-progress ordering. More generally, a seed whose influence is already in the current closed state can be omitted permanently: closure idempotence makes it irrelevant to all later activation, and removing an edge cannot hurt graphic independence. □

This theorem is the point at which full exceptional-row orders disappear. We need neither an order of all b rows nor a partition of demands sharing a seed. A single seed's entire influence mask is inserted at once.

4 Mandatory choices as a generic contraction

4.1 The graph representation

Orient the nonloop edges of D arbitrarily. Let $V_D = (v_e)_{e \in E}$ be its reduced incidence matrix over a prime field $\mathbb{F}_p$, with the row of the boundary root removed. It has r rows. A loop has the zero column. Over every field, the columns indexed by a set of edges are linearly independent if and only if that edge set is a forest. For a forest, repeatedly removing a nonroot leaf produces a square minor with determinant 1 or -1. A cycle, including a loop or a two-edge parallel cycle, gives a column dependence. This also explains why characteristic two is harmless for the *graph representation*.

Introduce an independent indeterminate $z_{i,e}$ for each $e \in U_i$, and set

$$y_i = \sum_{e \in U_i} z_{i,e} v_e, \qquad Y = [y_1 \ \cdots \ y_q]. \tag{10}$$

These columns initially live over the rational function field $\mathbb{F}_p(\mathbf{z})$.

Lemma 4.1 (Generic representative contraction)**.** *For any set X of k edge labels, the following are equivalent:*

(i) $[Y, V_{D,X}]$ *has rank* $q+k$ *over* $\mathbb{F}_p(\mathbf{z})$*;*

(ii) one can choose representatives $u_i \in U_i$*, distinct from one another and from* X*, for which* $X \cup \{u_i : 1 \le i \le q\}$ *is a forest.*

If a specialization of the $z_{i,e}$ has rank $q+k$, condition (ii) holds for the original graph, without any probabilistic qualification.

Proof. Expand any $(q+k)$-row minor in the first q columns. Its terms are indexed by tuples $(u_1, \dots, u_q)$, with monomial $\prod_i z_{i,u_i}$ and coefficient equal to the corresponding graph-incidence minor with the columns X appended. Different tuples give different monomials because the variables retain both the color and the edge label. Thus a minor is a nonzero polynomial precisely when at least one of these coefficients is nonzero.

A nonzero coefficient implies a forest of $q+k$ distinct edges. Conversely, a forest of representatives together with X has a graph minor equal to ± 1. Its monomial coefficient cannot cancel with any other tuple, and remains nonzero over every prime field. Finally, a polynomial that evaluates to a nonzero value is not the zero polynomial. Specialization may destroy independence but cannot create an independent set that lacks genuine representatives. □

If no independent representatives exist even for $X = \varnothing$, the local problem is negative. A polynomial graphic/partition matroid-intersection prepass detects this and can provide a Rado obstruction: a subcollection $J \subseteq \{1, \dots, q\}$ with

$$|J| > \operatorname{rank}_D \left(\bigcup_{i \in J} U_i \right). \tag{11}$$

The delivered checker verifies this inequality directly from graph components. This classical prepass is an optimization and an exact-negative route; it is not the new single-exponential argument.

4.2 Specialize, quotient, and reduce the rank

Choose every $z_{i,e}$ uniformly and independently in $\mathbb{F}_p$. If the sampled Y has rank less than q, abandon this round. Contracting fewer than q independent columns would forget mandatory constraints and is not allowed. Otherwise compute a linear quotient map

$$Q : \mathbb{F}_p^r \longrightarrow \mathbb{F}_p^{r-q}, \qquad \ker Q = \langle Y \rangle.$$

Gaussian elimination constructs Q: reduce a vector by a normalized echelon basis of Y and retain its free coordinates. Put $d = \min\{b, r-q\}$ and independently choose a uniform random matrix $P \in \mathbb{F}_p^{d \times (r-q)}$. The small representation used by the search is

$$a_e = PQv_e \in \mathbb{F}_p^d. \tag{12}$$

A set independent in this representation was independent in the quotient before projection, and therefore has genuine mandatory representatives by Lemma 4.1. Rank reduction only loses potential witnesses.

Remark 4.2 (Do not apply the field to geometric counts)**.** The field acts on V_D, not on N. For example, $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ is a stopping matrix, but its reduction modulo two is a permutation matrix and peels completely. Replacing unsigned incidences by residues would change the problem. The implementation keeps all heavy prerequisites even in the fully deterministic characteristic-two search.

4.3 A witness can be unmixed deterministically

Proposition 4.3 (Algebraic unmixing)**.** *Suppose sampled Y and a set X have* $\operatorname{rank}[Y, V_{D,X}] = q + |X|$. *Actual independent representatives can be recovered in polynomial time using only field-rank tests and the lists U_i.*

Proof. Maintain the independent list consisting of the q current columns and $V_{D,X}$. At step i, its ith column is still the original sampled linear combination $y_i \in \langle v_e : e \in U_i \rangle$. Its exterior product with the other columns is nonzero. Multilinearity implies that replacing y_i by at least one v_e, $e \in U_i$, leaves a nonzero exterior product. Scan the candidates with rank tests, choose such an edge, and keep that replacement. After q steps every mandatory column is an actual edge. Independence guarantees that no edge was repeated or already belonged to X. The number and dimensions of the rank tests are polynomial in the explicit input size. □

The current producer instead invokes its already tested graphic/partition oracle with X contracted, to recover representatives. Proposition 4.3 is a proved alternative, not an additional executed recovery path in the package. It also shows that mandatory matroid intersection is not logically necessary for the randomized positive-search theorem: the exact rank- obstruction prepass may be omitted and all positive witnesses can be recovered by linear algebra.

5 Exterior representative families over closed states

5.1 The representative property

For an ordered independent k-tuple $X = (e_1, \dots, e_k)$ in the small representation, write

$$w_X = a_{e_1} \wedge \cdots \wedge a_{e_k} \in \bigwedge^k \mathbb{F}_p^d.$$

This space has dimension $D_k = \binom{d}{k}$. The coordinates of w_X are the $k \times k$ minors of the matrix with columns a_{e_i}. In particular, $w_X \neq 0$ exactly when X is independent.

Lemma 5.1 (A spanning subfamily preserves extensions)**.** *Let $\mathcal{F}$ be a family of independent k-sets having the same closed activation state S. Choose actual members $\widehat{\mathcal{F}} \subseteq \mathcal{F}$ whose exterior vectors form a basis of the span of all $\{w_X : X \in \mathcal{F}\}$. Then $|\widehat{\mathcal{F}}| \leq \binom{d}{k}$. For every future edge set Z with $|Z| \leq d - k$, if some $X \in \mathcal{F}$ has $X \cup Z$ independent, then some $\widehat{X} \in \widehat{\mathcal{F}}$ does too. Starting from X or $\widehat{X}$, the future activation caused by the same Z is identical.*

Proof. The map $w \mapsto w \wedge w_Z$ is linear. If it is nonzero on any vector in the span, it is nonzero on at least one vector of a spanning basis. A nonzero product guarantees both independence and disjointness from Z. The activation state records everything a future seed can affect:

$$\operatorname{cl}\left(S \cup \bigcup_{e \in Z} H(e)\right).$$

It does not depend on the particular set used to reach S. □

This is the standard linear-matroid representative-family mechanism [9]. Its application here is valid because Theorem 3.2 has separated all future activation information from the identity of the already selected seed edges.

5.2 The dynamic program

At state (S, k) retain a basis of exterior vectors of actual k-seed witnesses whose closed state is S. Start with the empty witness and the scalar exterior vector 1 at $(\mathrm{cl}(\varnothing), 0)$. For every retained witness X and every edge e, form

$$S' = \mathrm{cl}(S \cup H(e)), \qquad w' = w_X \wedge a_e. \tag{13}$$

Ignore the transition if $S' = S$ or $w' = 0$. Otherwise insert its actual witness $X \cup \{e\}$ into the basis at $(S', k+1)$ if its exterior vector is not already in that span. If $S' = B$, return the witness immediately. Process states by increasing k; only $0 \le k \le d$ can occur.

Theorem 5.2 (Completeness in a supplied representation)**.** *For any fixed vectors $a_e \in \mathbb{F}_p^d$, the dynamic program returns a witness if and only if some independent edge set X satisfies* $\mathrm{cl}(\bigcup_{e \in X} H(e)) = B$.

Proof. Every retained witness is actual, independent, and has its recorded closed state, by construction. This proves soundness.

For completeness, choose a successful independent witness and remove redundant seeds as in Theorem 3.3. Its seeds can be added in strict-progress order. When an intermediate witness is discarded in favor of a basis, Lemma 5.1 preserves an extension by the remaining ordered seed set. The replacement has the same state, and the nonzero exterior product with all remaining seeds guarantees that every prefix of that extension remains independent. The next activation state is therefore unchanged. Induct on the number of remaining additions until the full boundary is reached. This argument also covers replacements of a witness that already arose as a previous representative replacement. □

5.3 A representation invariant that matters in code

Gaussian elimination of a candidate exterior vector can produce a linear combination that is not the exterior vector of *any* actual seed set. Consequently an eliminated basis row must not be treated as a new witness. The implementation stores two objects separately: normalized echelon rows for span tests, and each retained original decomposable vector together with its actual ordered edge tuple. Transitions extend only the latter. A dedicated regression test distinguishes the raw vector from its eliminated row.

The literal-set ablation also retains actual witnesses. It removes repeated *sets* reached through different orders, but does not apply exterior span compression. The measured saving is therefore not merely suppression of different permutations of an identical edge set.

6 The single-exponential search theorem

6.1 Why only one witness has to survive randomness

We use the elementary polynomial zero bound: a nonzero polynomial of total degree at most t over $\mathbb{F}_p$, evaluated at independent uniform field values, vanishes with probability at most t/p. For completeness, write it as a polynomial in its last variable, with leading coefficient of degree at most $t - s$ and last-variable degree s. The leading coefficient vanishes with probability at most $(t-s)/p$ by induction; otherwise there are at most s roots in the last coordinate. This proves the bound. It is the usual Schwartz–Zippel argument [11].

Lemma 6.1 (Fixed-witness survival)**.** *Fix a feasible k-seed witness X_* in Theorem 3.3. One random specialization and projection retain its independence with probability at least*

$$1 - \frac{q+k}{p}. \tag{14}$$

The witness is fixed before the random variables of this round are sampled.

Proof. By Lemma 4.1, some $(q+k)$-minor of $[Y, V_{D,X_*}]$ is a nonzero polynomial. Its degree is at most q, because only the mandatory columns are random. Thus this augmented matrix loses rank with probability at most q/p. On the complementary event Y has rank q, so the round is not abandoned, and the quotient images of X_* are independent.

Condition on any such specialization of Y. The independent quotient images have a k-dimensional span. Since $k \le d$, there exists a linear map to $\mathbb{F}_p^d$ that preserves them. Accordingly some k-minor of the random projected columns is a nonzero polynomial of degree at most k in the independent entries of P. Its failure probability is at most k/p. The conditional bound holds for every successful first-stage specialization. Adding the two failure probabilities proves (14). □

No union bound over every feasible seed set is needed. On a positive input, one fixed successful witness suffices. Once the small representation retains that witness, Theorem 5.2 completes its search deterministically. Requiring a simultaneously faithful representation of all $m^{O(b)}$ possible seed sets would be an unnecessary extra demand.

Theorem 6.2 (One-sided single-exponential selector)**.** *For the fixed local problem of Section 2, there is an algorithm with these properties:*

(i) Every reported success has a valid spanning tree and complete peeling certificate, checked deterministically from the original graph and matrix.

(ii) If a successful tree exists, λ uncapped independent rounds find one with probability at least $1 - 2^{-\lambda}$.

(iii) Its bit running time is $2^{O(b)}\,\mathrm{poly}(\Lambda, \lambda)$. One round admits (1) as a conservative field-operation bound, with $d = \min\{b, r-q\}$ when mandatory representatives are feasible.

(iv) Its field-element storage can be bounded by $2^b 4^d\,\mathrm{poly}(\Lambda) \le 8^b\,\mathrm{poly}(\Lambda)$.

On an unsuccessful search the general algorithm reports `UNKNOWN`*, not a negative topological answer. An optional exact rank-obstruction route may certify local infeasibility sooner.*

Proof. First handle the empty input and any automatically complete closure. The mandatory prepass either gives the verified obstruction (11), or ensures $q \le r$. Alternatively it can be omitted and rank-deficient mandatory samples can simply be abandoned.

Choose a prime $p > 2(q+b)$ with $p = O(q+b+1)$; elementary prime bounds and trial division suffice in polynomial time for these explicitly bounded parameters. The implementation takes the first prime at least $\max\{101, 2(q+b)+1\}$. Sample Y, check its rank, quotient, and project as in Section 4. Search by Theorem 5.2. If X is found, sampled independence implies genuine extendibility by Lemma 4.1. Recover mandatory representatives by the polynomial forest oracle or Proposition 4.3; extend their union with X to a tree. Then replay the actual matrix peeling certificate. All these steps are deterministic after the sample, and no field outcome can cause a false positive.

On a positive input fix the witness furnished by Theorem 3.3, independently of the sampling. By Lemma 6.1 a round misses it with probability less than $1/2$. Conditional on retaining it, the dynamic program succeeds. Independent repetition gives the stated miss bound.

For the complexity, each mask and cardinality state retains at most $D_k = \binom{d}{k}$ witnesses. There are at most 2^b masks. Therefore the number of candidate edge extensions is at most

$$m\,2^b \sum_{k=0}^{d} D_k = m\,2^{b+d}.$$

An exterior extension costs $O(dD_k)$ arithmetic operations in a direct coordinate implementation. A straightforward insertion into a basis in dimension D_{k+1} costs $O(D_{k+1}^2)$. Since

$$\sum_{k=0}^{d-1} D_k D_{k+1}^2 \le \left(\sum_k D_k\right)\left(\max_k D_k\right)^2 \le 8^d,$$

the claimed time bound follows. Echelon bases and raw witness vectors occupy at most a polynomial factor times

$$2^b \sum_{k=0}^{d} \binom{d}{k}^2 = 2^b \binom{2d}{d} \le 2^b 4^d$$

field elements. Influence preparation, closure saturation, graph operations, quotients, and certificate output are polynomial factors in Λ. Field elements have $O(\log(q+b+2))$ bits. Standard exact finite-field arithmetic gives the stated bit bound. The research implementation uses sparse dictionaries; even conservative dictionary overhead is absorbed in $2^{O(b)}\,\mathrm{poly}(\Lambda,\lambda)$, rather than being asserted to have a particular machine-level base of sixteen. □

The algorithm's probability statement assumes fresh independent uniform samples and enough resources to complete the claimed rounds. A fixed pseudorandom seed is useful for regression reproduction, not a derandomization. A boundary or transition cap may cause `UNKNOWN` even on a positive input, so a capped call is not automatically entitled to the uncapped miss guarantee.

7 Deterministic low-rank decisions and frontier certificates

7.1 A class with no randomness at all

Theorem 7.1 (Faithful low-rank selection)**.** *Suppose $q = 0$ and $r \le b$. Successful dual-tree selection is decidable deterministically in $2^{O(b)}\,\mathrm{poly}(\Lambda)$ time. A positive answer has the same tree-and-peeling certificate as before. A negative answer has a frontier-span certificate of at most $4^b\,\mathrm{poly}(\Lambda)$ bits, verifiable independently in single-exponential time.*

Proof. Use the original reduced binary incidence columns $v_e \in \mathbb{F}_2^r$. They represent graphic independence exactly; no mandatory contraction or rank projection is necessary. Run the dynamic program with $d = r$. Its soundness and completeness follow from Theorem 5.2 and the small seed theorem. A negative answer can retain the successful basis insertions at every state. There are at most $2^b \sum_k \binom{r}{k} \le 2^{b+r}$ actual witnesses, each with at most r edge labels. The certificate and verification claims follow from the explicit construction below. □

7.2 What the negative packet contains

A frontier packet contains an input digest and, for every retained state (S, k), a list of actual k-edge seed witnesses whose exterior vectors are a basis for that state. It contains no alleged field-operation

transcript and no claim that a reduced linear combination is an actual witness. The complete boundary state B must not occur.

An independent verifier performs the following checks.

1. Recompute B and the mandatory components from the supplied matrix; verify $q = 0$ and $r \leq b$. Check input binding, unique state keys, label ranges, and the absence of a full-boundary state.
2. For each alleged witness X, recompute its exterior coordinates as binary determinants of graph-incidence minors. Check that the vector is nonzero and that literal matrix peeling from X has precisely the claimed exceptional projection. Verify that each state's listed vectors are linearly independent.
3. Check that the empty exterior vector 1 is in the initial state's span. For each state witness X and each edge e giving strict activation progress, recompute $w_{X\cup\{e\}}$. If nonzero, require it to belong to the listed span at the appropriate next state. At $k = r$ every further edge extension is dependent, so there is nothing to check.

The supplied implementation computes activation in this checker by literal peeling, not by importing the producer's Horn-closure code. It computes minors directly, not by importing the producer's wedge-extension routine.

Lemma 7.2 (Frontier coverage proves local nonexistence)**.** *If these checks pass, no successful tree exists for the supplied local input.*

Proof. The initial state spans the empty witness. Induct on a strict-progress seed sequence. Suppose the exterior vector of its current prefix is in the span of the listed state basis. Wedge multiplication by the next edge is linear, and its next activation state depends only on the current state and that edge. The transition checks therefore place the next prefix vector in the listed next-state span. If a successful independent seed set existed, choose a strict-progress ordering as in Theorem 3.3. Its final vector would be nonzero in a full-boundary state, contrary to the packet's absence of such a state. A successful tree supplies such a seed set. □

This is an exact local negative certificate, unlike a failed random projection. It must not be extended silently to $q > 0$, where a specialized mandatory contraction can lose witnesses, or to $r > b$, where a sampled projection may do so. The verifier explicitly rejects those unsupported regimes. Nor is a valid frontier packet a certificate of knottedness.

8 Necessary parity checks for geometric source data

The abstract selector deliberately allows matrices and graphs that may not come from a three-manifold. There is nevertheless a useful inexpensive source-consistency condition, separate from the selection theorem.

Proposition 8.1 (Dual-cycle parity)**.** *Suppose N gives* exact *unsigned one/two-handle intersection counts, and D gives the actual two/three-handle endpoint occurrences, in a handle chain complex with the stated lower forest contraction. Then*

$$V_D(N \bmod 2) = 0 \quad \textit{over } \mathbb{F}_2. \tag{15}$$

Equivalently, for each column of N, the dual edges on which the count is odd have even degree at every dual vertex.

Proof. Over $\mathbb{F}_2$ signs disappear. In handle bases, ∂_2 is the transpose of the unsigned incidence matrix modulo two, and ∂_3 is the transpose of the reduced dual incidence matrix. The chain identity $\partial_2\partial_3 = 0$ gives (15) after transposing. The equation at the omitted boundary root follows from the sum of the other incidence rows. A loop contributes twice and therefore contributes zero modulo two. □

This is a specialization of the chain identity, not a new general theorem about chain complexes. It is relevant because it detects a class of abstract fixtures that must not be advertised as genuine geometric sources.

The test cannot be applied directly to entries that were merely capped at two. An entry recorded as "at least two" may have either actual parity. There is an exact polynomial criterion for whether the *saturated categories* admit any compatible parity lift.

Theorem 8.2 (Parity lifting for saturated columns)**.** *For one saturated column, let U be its unit edges and H its heavy edges. Let O be the vertices of odd degree in the dual subgraph with edge set U. A choice of parities for the edges of H makes U together with the odd heavy edges Eulerian if and only if*

$$|O \cap K| \text{ is even for every connected component } K \text{ of } D[H]. \tag{16}$$

When feasible, a spanning-forest leaf sweep constructs the lift. All heavy entries can be realized as either 2 *or* 3 *for this parity purpose.*

Proof. A heavy-edge subgraph has an even number of odd-degree vertices in each heavy connected component, proving necessity. For sufficiency choose a spanning forest of $D[H]$. Root each component. Process its nonroot vertices from leaves toward the root; include a parent edge precisely when its child currently has odd demand, then toggle that demand at the parent. Only the root can retain demand, and (16) makes its final demand even. The chosen heavy edges have boundary O, so their symmetric sum with U is Eulerian. Assign count three to chosen heavy edges and count two to the others. Loops need not be chosen. □

The implementation supplies both diagnostics in `integration/parity_audit.py`. Neither authenticates source charts or constructs a manifold. A parity-compatible lift is not permission to change an already authenticated exact count. The maintained five-row matrix passes the exact parity test. The nonlinear choice family below also passes it; its three-dimensional realization remains unproved.

9 The maintained loop example and the integration seam

For the source with one interior dual vertex, the maintained matrix on columns with original labels $(3, 5, 6, 7, 8)$ is [4]

$$N = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

Rows $0, 1, 2, 3$ are parallel root-to-interior edges, while row 4 is an interior loop. Deleting row 0 leaves totals $(2, 3, 2, 3, 2)$, hence a stopping set. Here $b = 5$, $q = 0$, and $r = 1$, so Theorem 7.1 applies. The new search selects row 1 after two candidate transitions. Its local peeling schedule, in normalized column labels, is

$$(3, 0), \quad (2, 4), \quad (4, 1), \quad (0, 2).$$

In original column labels this is

$$(3,3), \quad (2,8), \quad (4,5), \quad (0,6).$$

The source's previously recorded alternative schedule used column 7 at the third step rather than column 5. Both are legal matrix schedules. The new local schedule consequently leaves a different column representative; one must not copy stale downstream port words or a prior basis-dependent certificate on the assumption that the chosen tree fixes the entire proof.

The delivered adapter translates only the verified tree into a native two/three reduction proposal. For a normalized interior identifier 37, its cancellation list is the single record

```
{two_handle: 1, three_handle: 37}.
```

It preserves the complete endpoint list, including the repeated endpoint of row 4, and records surviving source row IDs $(0,2,3,4)$. Native replay is responsible for renumbering those survivors. The adapter checks that its input graph really matches the supplied endpoint records, so the local certificate cannot silently be applied to another graph.

The maintained target is `fastunknot.core_three_handle_reduction_verify.replay_core_three_handles`. The required schema is `core-three-handle-reduction-v1`, with fields `dual_endpoints`, `cancellations`, and `surviving_two_handles`. Its current source checker was read at the pinned commit. The adapter's record normalization, source-ID mapping, mismatch refusals, and proposal were executed in seven tests. Full native geometry, Regina cuts, bundle-port replay, and end-to-end unknot recognition were *not* executed in this environment. These remain mandatory integration tests, not results inferred from the local example.

10 Implementation, resource semantics, and certificates

10.1 Modules and independent paths

The new package uses only the Python standard library. It was executed with CPython 3.13.5 on the Linux environment recorded in the result files. `model.py` validates the explicit graph and unsigned matrix; `closure.py` builds reachability and the exceptional closure; `matroid.py` implements a simple disjoint-color graphic/partition oracle; `algebra.py` performs finite-field and exterior operations; `solve.py` orchestrates the routes; and `verify.py` imports none of the producer, graph optimizer, closure, or exterior modules.

The polynomial preprocessing is intentionally straightforward. Low reachability uses bit masks and transitive closure rather than an optimized SCC/condensation implementation. The matrix producer uses a dense deterministic peeling scan. Neither choice is claimed to improve the maintained sparse peeler; the research target is the exponential *selection* dependence. The small-parameter method belongs behind a cheap incumbent attempt, as in the supplied default.

The positive certificate consists of an input digest, a spanning-tree edge list, its boundary-outward dual order, and a complete peeling schedule. A simple independent checker verifies the tree, dual moves, private columns, and full coverage. No random seed, field modulus, Gaussian transcript, or representative-family calculation is trusted to establish the answer. A SHA-256 digest prevents accidental input substitution; it is not a proof that an input came from a triangulation.

10.2 The three statuses

The public interface has three meanings. `SUCCESS` provides a verified local positive certificate. `BLOCKED_EXACT` provides either a verified mandatory-rank obstruction or a verified faithful low-rank frontier. `UNKNOWN` means that no such certificate was obtained within the supported route and resources. In particular a failed sample, an exhausted round allowance, an exceptional boundary cap, or a transition cap does not become a local negative answer. Even `BLOCKED_EXACT` is not a topological negative verdict.

The default caps are $b \leq 20$ and one million considered transitions across sampling rounds, with eight requested rounds. They are practical research safeguards, not the hypotheses of an unconditional guarantee for every positive input. Passing `None` as the corresponding API arguments runs the uncapped algorithm in the theorem. The callback propagates interruption exceptions; it does not translate them to a verdict. Source preparation and polynomial preprocessing are not bounded merely by a transition counter.

The validator rejects an impossible gigantic vertex count before allocating vertex arrays, using $|V| \leq m + 1$ for a connected explicitly listed graph. Empty-row input does not expand a potentially large declared count of zero columns. A 2^{16384} matrix coefficient is transported in hexadecimal and bound exactly by the input digest without decimal-expansion limits. The capability is tested, not presented as a benchmark of native normal-sheet multiplicity processing.

10.3 Reproduction commands

From the package root:

```
python -m unittest discover -s tests -v
python -m experiments.validate
python -m experiments.benchmark
python -m activation_forest examples/repository_loop.json --no-incumbent
python -m activation_forest examples/generic_positive.json --seed 1210
python -m activation_forest examples/repository_loop.json \
  --verify results/repository_loop_answer.json
cd article && sh build.sh
```

Fixed seeds in the archived experiments make their finite executions reproducible. Production use without `-seed` requests fresh system randomness. The old selector is vendored unchanged with its license and provenance solely for comparative reproduction; it is not counted as new implementation code.

11 Executed validation and measured tradeoffs

11.1 A ledger that distinguishes unknowns from negatives

The tests and audit scripts were executed, rather than only supplied. `results/validation.json` contains environment details, counters, seeds, and the driver hash. The following table summarizes distinct audit loops; unit-test assertions are not silently added to the input-case count.

Audit	Trials	Positive	Exact negative	Unknown
Exhaustive $4imes2$	6,561	2,586	2,555	1,420
Random matrix/graph	1,500	531	836	133
Planted generic positives	300	300	0	0

Additional algebraic audit	Queries or samples
Forced matroid extensions	16,038
Horn closure versus literal peeling	2,369
Sampled-independence soundness	1,951
Projected representative DP versus literal sets	255
Complete small-field sample spaces	503

The exhaustive matrix audit fixes a three-vertex four-edge dual multigraph with a parallel edge and visits every one of the $3^8 = 6561$ four-by-two matrices with entries in $\{0, 1, 2\}$. The random audit varies connected multigraphs, including loops, and dense matrices. Both compare with literal enumeration of every spanning tree. All 3,117 positive inputs in these two sets receive positive certificates. Of their 4,944 negative inputs, 3,391 receive exact local certificates and 1,553 remain `UNKNOWN`. Those inconclusive outputs are not counted as negative decisions or as proofs of full local completeness.

A further 300 planted positive trials use cycle dual graphs, $3 \le b \le 7$, and $1 \le q \le 6$. All succeed, requiring 318 total sampled rounds. These abstract families exercise the mandatory contraction and random rank reduction, rather than only the deterministic $q = 0$ route. They are not asserted to be geometric source matrices.

The 71 unit tests include positive and negative certificate corruption, producer-disabled replay, bad random samples, resource caps, interruption, loops and repeated endpoints, large binary counts, the record-level adapter, and the exact versus saturated parity distinction. The matroid-intersection audit compares 16,038 forced-extension queries against literal representative tuples. The projected-search audit compares the new dynamic program with literal independent-seed enumeration in exactly the same sampled representation; it does not confuse a sampled miss with an error in that representation's search.

11.2 An exact finite-field probability check

Take the triangle with edge vectors v_0, v_1, v_2, one mandatory class $\{0, 1\}$, and the fixed additional witness $\{2\}$. Sampling $y = z_0v_0 + z_1v_1$ and a one-dimensional quotient projection uses three field samples. The quotient witness survives the first stage with probability $1 - 1/p$, and the nonzero scalar projection survives with the same conditional probability. Thus the exact miss probability is

$$1 - (1 - 1/p)^2 = \frac{2p - 1}{p^2} \le \frac{2}{p}. \tag{17}$$

The script enumerates all 503 samples over $p = 2, 3, 5, 7$. The observed miss counts are respectively 6/8, 15/27, 45/125, and 91/343, exactly as in (17). This is a complete finite sample-space check, not a statistical estimate. Small fields can have substantial miss probability; that is why the theorem's constant-success choice is $p > 2(q + b)$.

11.3 A nonlinear family with exact witness-count separation

For $g \ge 1$, let the dual graph have root 0 and g leaves. On arm i place three parallel edges, in input order c_i, a_i, b_i. Add a loop z at the root. Introduce a column with unit entries on a_i, b_i, and a column

with entries 1 on c_i and 3 on a_i. Finally introduce one global column having entry 2 on every a_i, b_i and entry 1 on z. All unmentioned entries vanish. The use of 3 is intentional: it has the same activation category as 2, but makes this exact-count family satisfy Proposition 8.1.

There are $3g + 1$ rows and $2g + 1$ columns. The exceptional boundary consists of all a_i, b_i and z, so $b = 2g + 1$, $q = 0$, and $r = g$. There is no initial anchor. Each a_i, b_i is a low strongly connected pair, and deleting either also deletes c_i. The loop z can be deleted only when every good pair has disappeared, through the genuinely higher-support final column.

Proposition 11.1 (Choice-family counts)**.** *A spanning tree of this family succeeds exactly when it chooses a_i or b_i on each arm. There are 2^g successful trees out of 3^g. Before the first complete state is reached, the representative search retains $2^g - 1$ witness vectors, while deduplicated literal seed-set storage retains $3^g - 2^g$ witnesses. In the inspected predecessor's specified input and partition order, the first successful exceptional-order branch is its* Bell(g)*th matroid-intersection subproblem.*

Proof. A tree chooses exactly one of the three edges on each arm and cannot choose z. Choosing a_i or b_i activates the entire good pair and then c_i. Choosing only c_i leaves the good pair unavailable: the global column has no unit there, and neither low column can delete the first good row. Hence every arm must have a good choice, after which the global column deletes z. This proves the tree count.

For a proper subset J of activated arms, there are $2^{|J|}$ distinct independent useful seed sets, one good edge on each arm of J. In the binary star-incidence representation all these sets have the same nonzero exterior vector. One representative per J therefore suffices, giving $2^g - 1$ representatives. Literal deduplicated storage gives $\sum_{J \subsetneq [g]} 2^{|J|} = 3^g - 2^g$.

In the predecessor's first exceptional-row ordering the first row of each good pair imposes a demand with candidate set $\{a_i, b_i\}$; its partner is automatically reached, and z is reached at the end. The g demand sets are disjoint, so any partition merging two demands has an empty block candidate set. Only the discrete partition succeeds, and it is last in the audited restricted-growth partition order. Thus Bell(g) calls are made. This last assertion concerns the archived implementation order, not an inherent lower bound for every algorithm using exceptional orders. □

The graph rank and closure make this a useful ablation family, not a proposed hard family of knot diagrams. Satisfying the parity condition does not establish a three-dimensional realization. A specialized recognizer of its arm structure could solve it faster than either general-purpose method.

11.4 Paired local-query timings

Each timing begins with a validated matrix/graph instance and includes preprocessing, selection, certificate production, and one independent replay. The new producer already performs its replay before returning; a redundant second replay is not charged. The historical producer is followed by its independent checker. Source geometry construction, file I/O, and full knot recognition are excluded. Literal positive tree searches are also replayed. Literal negative enumeration has no separate compressed negative certificate and is reported as an exact-search control, not as an identical proof-format workflow.

The table gives median per-query milliseconds and the median of paired historical/new ratios. Five randomized-order pairs are retained per case. Submillisecond queries were batched; earlier unbatched data and earlier anchored-family controls are also retained and explicitly named. No case was removed merely because it favored the older or literal method.

Local input	Prior (ms)	New (ms)	Prior/new	Literal (ms)
Parallel negative, 3	0.1932	0.0786	2.61	0.0036
Parallel negative, 5	28.8081	0.1113	259	0.0068
Parallel negative, 6	642.6735	0.1358	4.78e+03	0.0080
Maintained loop matrix	1.1715	0.0733	17.4	0.0366
Nonlinear choice, 2 arms	0.0929	0.0868	1.07	0.0737
Nonlinear choice, 5 arms	1.0083	0.6783	1.5	8.0923
Generic positive	0.0596	0.0913	0.653	0.0202

The reconstructed maintained loop matrix benefits relative to the audited exceptional-order implementation, but literal testing of its very few trees is still faster. The generic positive example favors the prior implementation. The enormous historical/new ratio on the six-parallel-edge negative instance is particularly unrepresentative of generic difficulty: only six actual trees exist, and literal enumeration is much faster than either certificate-search framework. It measures avoidance of needless old order/partition branches, not evidence that difficult knot instances have been solved thousands of times faster. Odd-size members of that all-unit parallel family fail even the exact parity condition; the even-size member passes it, without proving geometric realizability.

For the nonlinear choice-family ablation, both arms perform the same activation preparation and discrete positive replay; only the compression of candidate seed families differs.

Arms	Literal witnesses	Representatives	Literal (ms)	New (ms)	Ratio
4	65	15	0.455	0.271	1.672
6	665	63	6.380	1.297	4.869
8	6,305	255	87.074	6.453	13.494

At eight arms, $6,305$ deduplicated literal witnesses are reduced to 255 representatives, as predicted by Proposition 11.1. All samples, operation counts, input hashes, source hashes, and same-algorithm A/A controls are in `results/benchmark.json`. These are finite Python measurements, not an empirical determination of an asymptotic exponent.

12 What this contributes to a quasi-polynomial recognizer

Corollary 12.1 (Conditional local quasi-polynomial regime)**.** *Let a recognition workflow on an n-crossing input make at most $L(n) = 2^{O((\log n)^2)}$ calls to this local selector. Suppose every call that must succeed has a successful forest, its encoded local input size is $2^{O((\log n)^2)}$, and its exceptional boundary satisfies $b = O((\log n)^2)$. Suppose all other charged work has the same quasi-polynomial bound. With fresh independent sampling per call and $\lambda = \lceil \log_2 L(n) + c \log_2 n \rceil$, the total charged local work is $n^{O(\log n)}$, and the probability that any required positive call is missed is at most n^{-c}.*

Proof. Substitute the hypotheses into Theorem 6.2 and multiply by $L(n)$. Both $2^{O(b)}$ and every fixed polynomial of the stated input size are $2^{O((\log n)^2)}$. The chosen λ is $O((\log n)^2)$ and adds only a polynomial factor in the exponent's scale. Conditional on the history before each call, that input is fixed and its fresh random variables have miss probability at most $2^{-\lambda}$. The union bound is therefore valid even for adaptively chosen later inputs. It gives $L(n)2^{-\lambda} \le n^{-c}$. □

The predecessor's $2^{O(b \log(b+1))}$ bound places the same local target, using only that estimate, in the smaller sufficient regime $b \log(b+1) = O((\log n)^2)$, for example $b = O((\log n)^2 / \log\log n)$. The new result removes this logarithmic penalty in the parameter allowance, at the stated randomization cost.

The corollary is *not* a complete unknot-recognition theorem. In particular, the required feasible-forest hypothesis cannot be replaced by "the local program usually succeeds." Some actual handlebody presentations can stall under restricted cancellation rules [1, 4], and some abstract inputs have no successful forest at all. A general `UNKNOWN` outcome on such an input has probability one, not a small failure probability. A correct integrated recognizer must continue through another verified route.

Proposition 12.2 (Exact fallback accounting on positive local inputs)**.** *Suppose a deterministic complete local fallback costs at most $F(\Lambda, b)$, and λ uncapped random rounds cost at most $\lambda K(\Lambda, b)$. Running those rounds first and invoking the fallback on `UNKNOWN` is always correct. On a positive local input its expected work is at most*

$$\lambda K(\Lambda, b) + 2^{-\lambda} F(\Lambda, b).$$

No corresponding small fallback probability follows for negative inputs.

Proof. Every early positive is independently valid. Otherwise the deterministic fallback decides the local problem. On a positive input the probability of reaching it is bounded by Theorem 6.2; on a negative input the random positive search cannot succeed, so it may always be reached. □

Similarly, repeating random rounds until success has expected fewer than two rounds on a promised positive instance, but need not terminate on a negative one. Such a promise procedure is not a total recognition algorithm. The finite-round public interface avoids that ambiguity.

The maintained supplied-cut representation has explicit combinatorial size bounds in terms of tetrahedra and ports [5]. Translating those bounds into a polylogarithmic exceptional boundary across all useful hierarchy stages remains a geometric theorem to be proved. Source surface discovery, admissible regularization, passive port transport, pattern testing, and restart charging still require their own bounds. Neither binary compression nor a small local positive certificate proves those bounds.

13 Further research questions and concrete next targets

13.1 Deterministic single-exponential mandatory contraction

Can the sampled contraction and rank reduction be replaced by a deterministic construction with $2^{O(b)}\,\mathrm{poly}(\Lambda)$ cost? A useful result must retain mandatory extendibility while reducing rank; choosing arbitrary mandatory representatives first can destroy a useful extra seed set. Fixing one pseudorandom seed is not a solution. An explicit hitting family tailored to the relevant graph minors, or a source-geometry restriction on their interaction, would be a genuine strengthening of Theorem 6.2. The unmixing proposition removes randomness from recovery, not from the existence-preserving search representation.

13.2 Source restrictions stronger than chain parity

Characterize saturated matrix/dual-graph pairs realizable by the maintained source charts. The parity-lift criterion is exact for its algebraic question but only necessary for geometry. Can intersection ordering, vertex-link planarity, bounded local source types, or boundary-port restrictions exclude the abstract hard instances used in previous work? A construction of a geometric realization of the nonlinear choice family would also clarify which of its phenomena are genuinely topological. Conversely, a rigorous nonrealizability proof would guide more meaningful benchmarks.

13.3 A uniform geometric bound on the activation boundary

The most direct route from the local theorem to the target complexity is a bound $b = O((\log n)^2)$ for all required positive stages, together with the other hypotheses of Corollary 12.1. It may be more flexible to bound the number of attainable closed masks rather than all 2^b masks, and to exploit the smaller quotient rank d. Any such bound must survive the actual hierarchy moves and not merely hold for one supplied cut.

13.4 Joint optimization of primal and dual forests

The one-handle basis is frozen throughout this article. Choosing a different zero/one forest can change both column supports and the exceptional boundary. Can a coupled selector trade primal changes against a smaller b while retaining independently replayable port charts? The graph-matroid part of the current proof does not automatically encode these basis changes. An end-to-end theorem would have to charge their effect on source occurrence words and on all later matrix construction.

13.5 More exact negative routes without full symbolic expansion

The frontier certificate is complete when the representation is known to be faithful. Can its coverage idea be lifted through a generic mandatory contraction without certifying every symbolic minor? Possible directions include rank-deficiency witnesses that explain all discarded mandatory choices at once, or a deterministic source-specific representation. A failed sampled frontier cannot be promoted to an original-input negative certificate by confidence rhetoric.

13.6 Fewer exceptional bits and stronger safe preprocessing

Rows used only as heavy prerequisites can dominate b. An alternative parameter based only on possible unit targets would need a compressed representation of their prerequisite states; simply dropping the heavy rows is invalid. Forced initial peeling and closure simplification may reduce the relevant boundary, but the original dual edges must remain available as forest resources even when their rows are already automatically active. Deleting them from the dual graph without justification can change the selection problem. A precise reduced interface for such preprocessing would be useful both theoretically and in implementation.

13.7 Weighted objectives and faster exterior computation

For integration, not every successful tree is equally good: subsequent port transport, word lengths, or proof size may matter. Weighted representative families preserve certain additive costs, but a genuine downstream objective need not be additive or determined by the current activation state. Identify an objective with a valid extension-compatible invariant before pruning by cost. Independently, implement sparse reachability, faster coordinate insertion, and low-rank specializations; the present general dictionary code is a research baseline rather than a tuned replacement for native sparse primitives.

13.8 Native replay, proof formalization, and adaptive sampling

The immediate integration experiment is to run the proposal on authenticated native source records, replay the source two/three certificate, and rerun all downstream checks without transporting a stale

local basis. The next formalization target is the closure reduction and representative-extension lemma, followed by the small deterministic frontier checker. Its lack of dependence on producer algebra should make its trusted kernel manageable. If samples are reused across adaptively generated local inputs, the simple fixed-witness probability proof no longer applies automatically; fresh samples or a stronger simultaneous guarantee must be provided.

14 Conclusion

The main result replaces a factorial/Bell-number search description by a closed-state, rank-bounded representative search. Its decisive invariant is that a successful dual tree contains at most b useful extra seeds beyond mandatory low-column representatives. Generic contraction handles the many mandatory choices, and exterior span compression preserves every relevant future independent extension. A fixed-witness probability argument avoids an unnecessary simultaneous-faithfulness demand.

This yields a one-sided single-exponential local search, a deterministic single-exponential low-rank subcase with checkable negative frontiers, and explicit source-parity diagnostics. The mathematics, executable kernels, independent checks, benchmarks, and integration boundaries are separated so that the contribution can be incorporated without treating an inconclusive local search as a knot verdict. The remaining global work is specific: source-faithful geometry, bounded activation complexity over the hierarchy, and complete charging of every required operator and restart.

A The classical mandatory-representative oracle

The prepass and current recovery implementation use graphic/partition matroid intersection, specialized to disjoint color classes U_i. A fixed forest X is contracted logically: the independence oracle tests whether $X \cup I$ is a forest. The partition constraint allows at most one selected edge from each mandatory class.

Starting with $I = \varnothing$, form the exchange digraph on selectable edges. An unselected edge is a source if it can be added to the forest $X \cup I$. An unselected edge is a sink if its color is unused. A selected edge y points to an unselected e when $X \cup (I - \{y\}) \cup \{e\}$ is a forest. An unselected nonsink points to the selected edge of its color. Toggle a shortest source-to-sink path. This is the classical common- independence augmentation; in the graphic specialization its exchanges can also be read directly from fundamental forest paths. The shortest-path condition prevents earlier exchange chords from invalidating the toggle. Iterating increases the common independent-set size until it is maximum. The standard matroid-intersection and Rado argument is documented in the audited predecessor [6]; this primitive is not asserted to be new.

For the uncontracted mandatory prepass, a failed augmentation yields a subcollection of colors satisfying (11). The implementation returns that subcollection, and the independent checker recomputes the mandatory components and graphic rank before accepting it. A maximum matching count alone would not be an accepted certificate. For recovery with fixed X, only actual representative edges are needed; their union with X and the resulting final tree are independently checked. The 16,038-query brute-force audit includes nonempty forced forests and empty or deficient color classes.

B Source audit and contribution ledger

The repository inspection is pinned, rather than silently following a changing main branch. The maintained files inspected include the directory README, the synthesis README, `synthesis/parallelity.tex`, `synthesis/bundle_boundary_kernel.tex`, `synthesis/core_three_handles.tex`, `synthesis/three_handles_results.tex`, and the complete current native three-handle replay function. The corresponding source paths and available Git blob identifiers are recorded in `SOURCE_AUDIT.json`.

The predecessor archive was read from the user's accessible Library and materialized for programmatic comparisons. Its article and code were used as a baseline; the vendored package retains its original license. `ProveIt_Nonlinear_Activation_Forests_2026-10-12.zip` and `ProveIt_Exceptional_Rows_2026-10-12.zip` were listed among the new incoming files but their contents were not obtained. The ordered-interface archive was inventoried, not used as a competing selector. Thus this study claims a demonstrated advance relative to the inspected matroidal article, not an exhaustive priority comparison against every incoming archive.

The ownership of the mathematical steps is also explicit. Private-column completeness, the dual-forest source charts, the pair-column interpretation, and the previous order/partition method come from the maintained and prior materials. Generic linear specialization, finite-field polynomial testing, graphic/partition intersection, and exterior representative families are standard tools. The proved contribution here is their closed-activation reduction and single-exponential combination, the faithful frontier certificate protocol, the application-specific parity audit, and the stated local quasi-polynomial consequences. The code, finite experiments, and limits on their interpretation are part of that contribution. Neither historical priority nor a completed global complexity theorem is asserted.

References

[1] V. Reshetnikov and ProveIt contributors. *Unknot Recognition: maintained project documentation.* `Topology/UnknotRecognition/README.md`, pinned commit `e0b2f802f336fcb51e9ec5f29174cfe28dfb9d39`, inspected 12 October 2026. https://github.com/VladimirReshetnikov/ProveIt.

[2] ProveIt contributors. *Complete rectangular cancellation by sparse column peeling.* `Topology/UnknotRecognition/synthesis/column_cancellation.tex`. The result is also developed in the inspected preceding research article [6] and referred to in the maintained kernel documentation.

[3] ProveIt contributors. *Removing source core three-handles relative to the bundle ports.* `Topology/UnknotRecognition/synthesis/core_three_handles.tex`, pinned commit `e0b2f802f336fcb51e9ec5f29174cfe28dfb9d39`.

[4] ProveIt contributors. *Source tests, dual-forest dependence, and three-handle results.* `Topology/UnknotRecognition/synthesis/three_handles_results.tex`, pinned commit `e0b2f802f336fcb51e9ec5f29174cfe28dfb9d39`.

[5] ProveIt contributors. *Parallelity geometry and the bundle-boundary cancellation kernel.* `Topology/UnknotRecognition/synthesis/parallelity.tex` and `Topology/UnknotRecognition/synthesis/bundle_boundary_kernel.tex`, pinned commit `e0b2f802f336fcb51e9ec5f29174cfe28dfb9d39`.

[6] ProveIt AI-assisted research contribution. *Matroidal Dual-Forest Selection for Unknot Recognition: Polynomial pair-column cancellation, exact deletion deficits, and a fixed-parameter*

higher-support kernel. 12 October 2026. Article and executable code in `docs/incoming/ProveIt_Matroidal_Dual_Forests_2026-10-12.zip`. Its maintained-source checkpoint is `c6d0ec8aa0b6308310ab6c081ba29d8bc363ede5`.

[7] Mathematical Institute, University of Oxford. *Marc Lackenby announces a new unknot recognition algorithm that runs in quasi-polynomial time*. 2021; inspected 12 October 2026. https://www.maths.ox.ac.uk/node/38304.

[8] M. Lackenby. *Incompressible surfaces, hierarchies and unknot recognition*. arXiv:2607.23350v1, 25 July 2026, especially Sections 9–10. https://arxiv.org/abs/2607.23350.

[9] F. V. Fomin, D. Lokshtanov, F. Panolan, and S. Saurabh. *Efficient Computation of Representative Sets with Applications in Parameterized and Exact Algorithms*. arXiv:1304.4626v4, 2016. https://arxiv.org/abs/1304.4626.

[10] R. van Bevern, O. Yu. Tsidulko, and P. Zschoche. *Representative families for matroid intersections, with applications to location, packing, and covering problems*. Discrete Applied Mathematics **298** (2021), 110–128. arXiv:1806.11527v4. https://arxiv.org/abs/1806.11527.

[11] J. T. Schwartz. *Fast probabilistic algorithms for verification of polynomial identities*. Journal of the ACM **27**(4) (1980), 701–717. DOI: 10.1145/322217.322225.